

MELANIE MASLOUB

La Jolla, California | (909) 485-7843 | melanie.masloub@gmail.com | LinkedIn:
www.linkedin.com/in/melanie-masloub-a0685437a

EDUCATION

University of California, San Diego, La Jolla, CA

Expected Graduation: Jun 2029

BS in NanoEngineering

Awards: IDEA Scholar, Boeing Scholarship, SEI (Summer Engineering Institute)

Redlands High School, Redlands, CA

Graduation: Jun 2025

Total GPA: 4.7447

Awards: Highest Honor Roll (2021-2025), CSF Life Member, AP Scholar with Distinction, Daisy Chain, Laurance H. Nowak Memorial Scholarship Recipient

RESEARCH EXPERIENCE

UCSD NanoBioElectronic Lab, *Undergraduate Research Assistant*, La Jolla, CA

Jan 2026 – Present

- Fabricating microneedle patches designed for sustained release drug delivery using nanofabrication techniques

PROJECTS

Nanoengineering and Technology Society, *Synthetic Nanofiber Skin (SyNS) Project Team Member* Oct 2025 – Present

- Synthesizing a synthetic nanofiber skin composed of a polycaprolactone base and a chitosan and gelatin crosslink
- Developing a structure that mimics that of the epidermis using the principles of biochemistry, nanofabrication, and technology

NANO 4, *Hybrid Graphene/Quantum Dot Photodetectors*

Sept 2025 – Dec 2025

- Fabricated and characterized a hybrid Graphene/Quantum Dot Based Photodetector using nanotechnology techniques like spin coating and graphene transfer

LEADERSHIP EXPERIENCE

RHS STEM Club, *Vice President and Treasurer*, Redlands, CA

Jan 2024 – Jun 2025

- Organized club fundraisers to purchase materials for STEM-focused activities
- Led club meetings for over 50 members, directing engaging STEM projects

A.K. Smiley Public Library, *Teen Leadership Team Leader*, Redlands, CA

Jan 2024 – May 2025

- Planned monthly events for teens, with approximately 30 attendees per event
- Brainstormed and orchestrated events and partnerships with local businesses
- Designed eye-catching social media advertisements for events using Canva

RHS Book Club, *Vice President and ICC Representative*, Redlands, CA

Aug 2023 – Jun 2025

- Led club meetings to discuss bi-monthly reads
- Coordinated with local library's youth librarian to buy and distribute physical book copies to our members for free
- Advocated for and implemented a democratic voting system to give members a say in our club reads
- Collaborated with cultural clubs on campus to highlight a diverse range of perspectives in our club reads

VOLUNTEER EXPERIENCE

A.K. Smiley Public Library, *Shelving/Youth Services Support Volunteer*, Redlands, CA

Sept 2023 – May 2025

- Mastered the Dewey Decimal System to shelve books in the youth department
- Prepared crafts for children, providing guidance and assistance
- Executed library closing procedures, including cleaning and organization

SKILLS

- *Hard Skills:* MATLAB, Microsoft 365, R, Onshape, Tinkercad, Google Workspace
- *Professional Skills:* Problem Solving, Creativity, Leadership, Communication, Teamwork